

1. A manufacturer of confectionary claims in their advert that 8 out of 10 children prefer the taste of their chocolate when compared to the next best-selling brand. Felix thinks that this claim is over-exaggerated, and decides to conduct an experiment at his local primary school. His hypotheses are:

$$H_0: p = 0.8$$

$$H_1: p < 0.8,$$

where p is the probability that a randomly chosen child will prefer the manufacturer's chocolate.

With the permission of their class teacher, 16 children try a small piece of each chocolate, and say which one they prefer. Felix then carries out a binomial hypothesis test at the 10% level of significance.

- a. Find the critical region for Felix's test.
 - b. Suppose that the manufacturer's claim is true. What is the probability that Felix will reject the null hypothesis?
 - c. Suppose that the manufacturer's claim is false, and that in fact just 6 out of 10 children prefer their chocolate. What is the probability that Felix will not reject the null hypothesis?
2. A local politician claims that 80% of people support her plans to redesign the town hall. A journalist thinks that fewer people than this actually support the plans, and decides to investigate. The journalist's hypotheses are:

$$H_0: p = 0.8$$

$$H_1: p < 0.8,$$

where p is the probability that a randomly chosen person from the town supports the plans.

The journalist interviews 96 local residents, and asks them their opinion on the plans to redesign the town hall. He then carries out a binomial hypothesis test at the 10% level of significance.

- a. Find the critical region for the journalist's test.
 - b. Suppose that the politician's claim is true. What is the probability that the journalist will reject the null hypothesis?
 - c. Suppose that the politician's claim is false, and that in fact just 60% of local residents support the plans. What is the probability that the journalist will not reject the null hypothesis?
3. [Reflect] Compare your answers to Questions 1 and 2.
 4. The manager of a coffee shop claims that two-thirds of people would rather have a latte than a cappuccino. Krya thinks that this claim has been completely made up, and sets out to investigate. She asks 24 people whether they would prefer a latte or a cappuccino, and carries out a hypothesis test at a 5% level of significance.
 - a. Write down suitable hypotheses for Krya's test, and determine the critical region.
 - b. If the manager of the coffee shop is correct, then find the probability that Krya will reject the null hypothesis.
 - c. If, in fact, people are equally likely to choose either a latte or a cappuccino, find the probability that Krya will reject the null hypothesis.
 5. In Question 4, suppose that Kyra had suspected that *fewer* than two-thirds of people would rather have a latte than a cappuccino. How would your answers have changed?

6. Artem has a biological model that predicts that three-quarters of people with a particular disease should have a particular variant of a certain gene. Rachel does not trust Artem's model and sets out to investigate Artem's claim for herself. She tests 238 people with the disease in question to determine how many of those people have that particular gene variant, and then conducts a hypothesis test at the 1% level of significance.
 - a. Write down suitable hypotheses for Rachel's test, and determine the critical region.
 - b. Suppose that Artem's claim is correct. Find the probability that Rachel's test will reject the null hypothesis.
 - c. Suppose that Artem's claim is not correct, and that 85% of people with the disease have the gene variant. Find the probability that Rachel's test will reject the null hypothesis.
7. In question 6, suppose that Rachel had suspected that *fewer* than three-quarters of people with the disease will have the gene variant. How would your answers to Question 6 have changed?
8. [BONUS] In the situation of Question 1, suppose that the manufacturer's claim is false, and that (unknown to Felix) three quarters of children prefer their chocolate. If Felix carries out a binomial hypothesis test at the 10% level of significance, then how large a sample does he need to take to have at least a 50% chance of getting the correct result?

[This will require trial and improvement.]

9. [BONUS] I claim that half of pupils studying Further Mathematics at Westminster will fully engage with this question. Dr Kalivas thinks that I am overly optimistic.

In fact, Dr Kalivas is correct, and the proportion of such pupils who fully engage with this question is p , where $0 \leq p < \frac{1}{2}$.

In an attempt to 'prove' me wrong, Dr Kalivas takes a sample of n Further Mathematicians and determines whether they have fully engaged with this question, and carries out a hypothesis test at the 5% level of significance.

Investigate the probability that Dr Kalivas (correctly) concludes that there is significant evidence to suggest that fewer than half of Further Mathematicians have fully engaged with this problem.

(You can safely assume that Dr Kalivas takes a random sample of pupils, asks questions so as to not introduce any bias, and carries out his test correctly.)